Isaac ROS 3D Pose Estimation

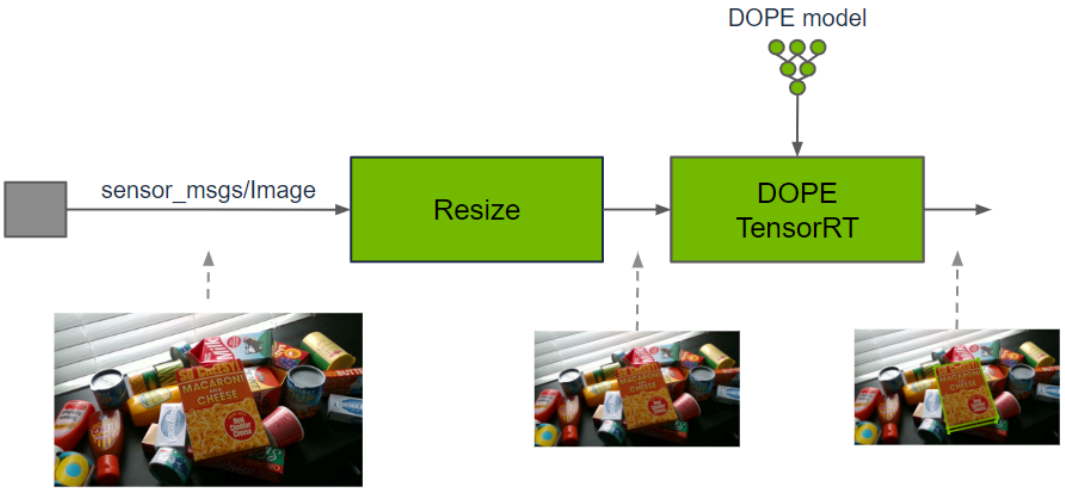
Isaac ROS 3D Pose Estimation
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Isaac ROS CenterPose official website link: https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_pose_estimation/isaac_ros_centerpose/index.html

Overview

Isaac ROS provides several object pose estimation solutions. CenterPose uses GPU-accelerated inference to predict the 3D position and pose of objects of known classes, and the results can be displayed together with the image and camera information.

Package	Feature
`isaac_ros_foundationpose`	Supports new objects without retraining
`isaac_ros_dope`	Pose estimation for known objects
`isaac_ros_centerpose`	Pose estimation that supports the TAO workflow



Nano Model Precision Selection

On the Jetson Orin Nano and under the current TensorRT environment, this case uses the FP32 engine: `/workspaces/isaac_ros-dev/isaac_ros_assets/centerpose_shoe_fp32.plan`. The companion script of this tutorial, `scripts/centerpose_launch.sh`, already specifies `centerpose_shoe.onnx` and this FP32 plan, and sets `force_engine_update:=False`; normal operation does not require moving files between model directories, and do not switch to FP16.

Before starting, the model files and the script selection can be confirmed:

```
ls -lh /workspaces/isaac_ros-dev/isaac_ros_assets/centerpose_shoe.onnx
/workspaces/isaac_ros-dev/isaac_ros_assets/centerpose_shoe_fp32.plan
rg -n 'model_file_path|engine_file_path|force_engine_update'
/workspaces/isaac_ros-dev/scripts/centerpose_launch.sh
```

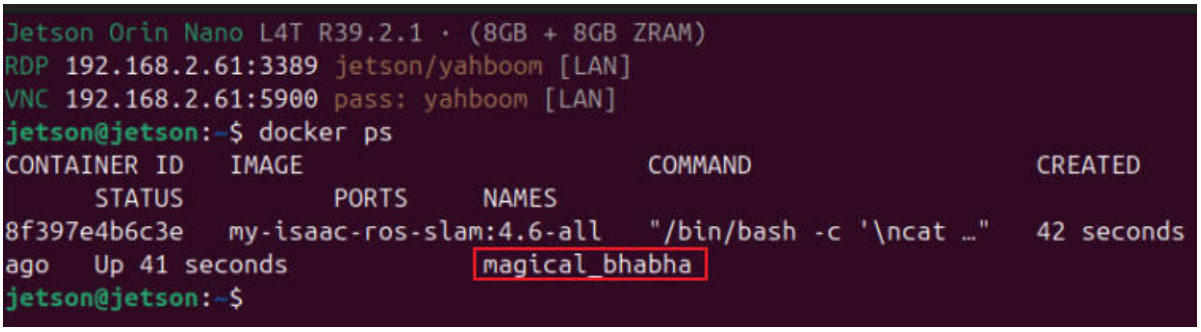
If the FP32 plan is missing, do not replace it with FP16 or force a rebuild of the engine; contact the course provider to obtain the model files that match this environment, and then run the case.

Quick Experience

First use the command below to start the container

```
./run_isaac.sh
```

Then confirm the container on the host with `docker ps`, and open container terminals with `docker exec -it <container name> bash`; each terminal should use the same ROS/FastDDS environment, for example:



Note: The terminals below all execute inside the container:

> After the power mode is set to MAXN, an overcurrent warning may pop up when running the case; ignore it.

Terminal A: Start inference:

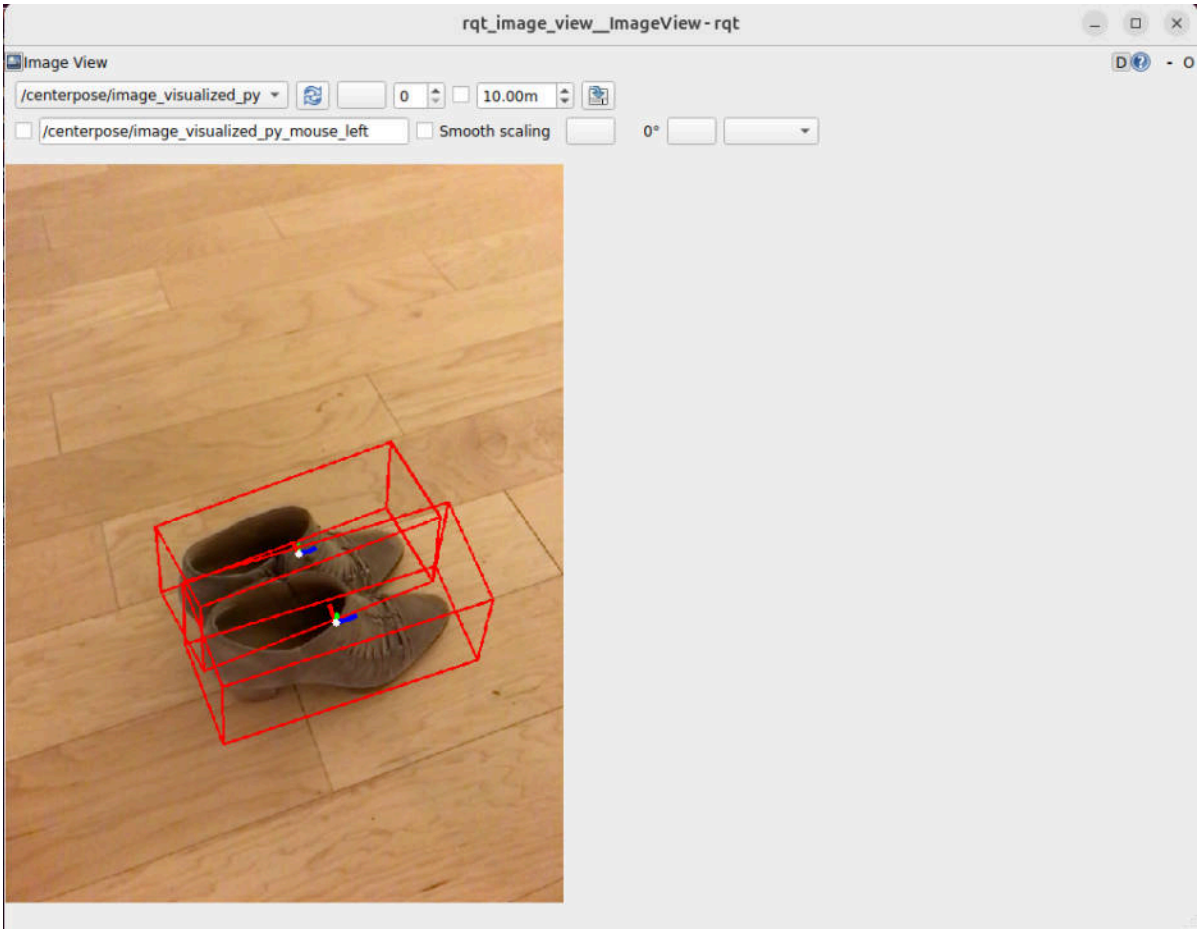
```
bash /workspaces/isaac_ros-dev/scripts/centerpose_launch.sh
```

Terminal B: Start the two bags required for inference and display:

```
bash /workspaces/isaac_ros-dev/scripts/centerpose_playback.sh
```

Terminal C: Start the Python pose box visualization and rqt:

```
bash /workspaces/isaac_ros-dev/scripts/centerpose_view.sh
```



Select `/centerpose/image_visualized_py` in Topic of rqt. To check the output, run:

```
ros2 topic info /centerpose/detections -v
ros2 topic echo /centerpose/detections --once
ros2 topic hz /centerpose/image_visualized_py
```